Hanxi (Gary) Chen

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EDUCATION

Cornell University — Ithaca, NY
Ph.D. in Computer Science, Programming Languages — Present
Advisor: Prof. Alexandra Silva, Prof. Andrew C. Myers

University of Pennsylvania, *Penn Engineering & The Wharton School* — Philadelphia, PA
M.S.E. in Computer and Information Science — 2025
B.S.E. in Computer Science, Mathematics minor — 2025
B.S. in Economics — 2025
Cumulative GPA: 3.97/4.00, *Summa Cum Laude*
Advisor: Prof. Steve Zdancewic, Prof. Sanjeev Khanna

PUBLICATIONS

Hanxi Chen, Noam Zilberstein, Andrew C. Myers, Alexandra Silva. Oblivious Probabilistic Outcome Logic: Verifying Probabilistic Programs with an Oblivious Adversary.

Calvin Beck, Hanxi Chen, Steve Zdancewic. Vellvm: Formalizing the Informal. The Eleventh International Workshop on Coq for Programming Languages (CoqPL), 2025.

Calvin Beck, Hanxi Chen, Steve Zdancewic. Vellvm: Formalizing the Informal LLVM (An Experience Report). Nasa Formal Methods: 17th International Symposium (NFM), 2025.

Calvin Beck, Irene Yoon, Hanxi Chen, Yannick Zakowski, Steve Zdancewic. A Two-Phase Infinite/Finite Low-Level Memory Model: Reconciling Integer–Pointer Casts, Finite Space, and undef at the LLVM IR Level of Abstraction. Proceedings of the ACM on Programming Languages, 8 (ICFP), 2024.

Manuscripts

Silei Ren, Suraaj Kanniwadi, Hanxi Chen, Andrew C. Myers. Type-Confusion Security.

Calvin Beck, Hanxi Chen, Steve Zdancewic. GenLLVM: Randomized Differential Testing for LLVM IR.

Hanxi Chen. Knapsack Problems with Precedence Constraints. Wharton Undergraduate Thesis, 2024. Supervised by Prof. Sanjeev Khanna.

Hanxi Chen. Random Differential Testing at the LLVM IR Level. CIS Undergraduate Thesis, 2024. Reviewed by Prof. Steve Zdancewic (advisor) and Prof. Benjamin C. Pierce.

RELEVANT RESEARCH EXPERIENCE

Research Assistant, *Cornell PL Group*, advised by Prof. Alexandra Silva and Prof. Andrew C. Myers — Sep 2025 –
Cornell University, Computer Science

- Conducting research on probabilistic concurrent outcome logic and probabilistic nondeterministic outcome logic under an oblivious scheduler, exploring applications in randomized distributed protocols, information flow systems, and machine learning optimization.
- Developing a Lean mechanization library for the logics, including a monadic semantics for probabilistic nondeterministic programs, a probabilistic assertion language, semantic and syntactic outcome triples, and new probability space constructions and automation using Mathlib and metaprogramming.

Research Assistant, *PLClub*, advised by Prof. Steve Zdancewic — Jun 2021 – Aug 2025
University of Pennsylvania, Computer and Information Science

- Contributed to the Verified LLVM (Vellvm) Project, formalizing LLVM IR semantics in Rocq. Formalized nontrivial lemmas such as serialization. Participated in the development of a low-level memory model in Rocq to facilitate pointer-integer casting.
- Developed a Rocq-based LLVM differential testing framework, as part of the Vellvm project, using QuickChick, capable of generating UB-free LLVM programs with nontrivial features like pointer manipulations. Reported novel bugs to the LLVM development team.
- Developed a program tracer using OCaml and Rocq for generating the dynamic trace via Vellvm interpreter. The tracer can generate annotated program trace, as well as well-formed LLVM IR program, for efficiently detecting bugs according to Vellvm's semantics.
- Developed a permutation library and proved the equivalence of seven variations of list permutations based on type class. Building a complete solver using MetaCoq for solving associativity-commutativity-unification on lists.
- Formalized the multiplicative, exponential first-order classical linear logic in Rocq with properties like cut elimination as a potential use case for the permutation library and a related formalization of opportunistically parallel, linear-resource programming language with asynchronous queries.
- Contributed to a newer version of Software Foundations, appeared at the Oregon Programming Language Summer School (OPLSS) at Boston University 2024, as part of the DeepSpec project.

Research Assistant, *Wharton Research Scholars*, advised by Prof. Sanjeev Khanna — Jan 2023 – Dec 2024
University of Pennsylvania, The Wharton School

- Researched and proved the existence of a pseudo-polynomial solution for precedence constraints knapsack problems with two different graph classes, union of three paths and one-directional bipartite poset with consecutive children.
- Gave a talk in Wharton Research Scholars seminar. Submitted a 48-page undergraduate technical report to Wharton Research Scholars Program in December 2024.

Research Assistant, *Penn Computational Intelligence Lab*, advised by Prof. Jing Li — Feb 2021 – Sep 2021
University of Pennsylvania, Electrical and Systems Engineering

- Co-designed and developed a novel simulator prototype, written in C++, that models the functionality of Intel's Optane Memory.
- Researched a C# solution in modifying PDFSplit, a PowerPoint add-in in Linux, to Microsoft.

TECHNICALS

RISC-V Pipelined Processor with cache – *SystemVerilog, Python* — Feb 2025 – May 2025
Keyword: RTL design, Formal Verification
University of Pennsylvania, *CIS 5710 Computer Organization and Design*, supervised by Prof. Joe Devietti

- Independently developed a fully functioning RISC-V pipelined processor project individually, including AXI-Lite-style caching, multi-cycle execution units, and basic branch prediction.
- Verified the processor using extensive custom testbenches, optimized the design for low LUT usage and timing performance, and synthesized it onto an FPGA to run real programs, including a Candy-Crush-style game.

OAT-Compiler: From C-like Lang to Assembly – *OCaml, C, LLVM, X86, Clang, Dune* — Feb 2024 – May 2024
Keyword: Code Generation, Static Analysis, Optimization
University of Pennsylvania, *CIS 7000 Compilers and Interpreters*, supervised by Prof. Steve Zdancewic

- Independently developed a workable full-stack compiler with linter, parser, frontend, and backend capable of compiling Oat, a C-like language with pointer manipulation and complex data structures, down to LLVM IR and X86.
- Validated with thousands of student test cases. The compiler contains a type system, analyses, and optimizations including mem2reg and dead-code elimination.

SQL-Interp: Haskell Library for SQL Interpretation and Parsing – *Haskell, MySQL, Cabal* — Sep 2023 – Dec 2023
Keyword: Functional Abstraction, Type Systems, Monadic Effects
University of Pennsylvania, *CIS 5520 Advanced Programming*, supervised by Prof. Stephanie Weirich

- Led a team in developing a model for MySQL syntax as well as a tested Haskell library for parsing, interpreting, optimizing, and pretty-printing SQL scripts, using GADTs, monad transformers, applicative parsers, and property-based testing.
- Formalized abstract datatypes covering most SQL features. Developed an interface for locally executing queries.

PennOS: Unix-like Operating System – *C, Valgrind* — Jan 2023 – May 2023
Keyword: Concurrency, Process Management, File Systems
University of Pennsylvania, *CIS 5480 Operating Systems*, supervised by Prof. Boon Thau Loo

- Led a team in designing a fault-tolerant, multi-thread operating system with kernel and file system components and no memory leaks.
- Co-developed kernel components including threads, signals, foreground/background execution, scheduler, API, and user-level functions, and integrated the file system.

HONORS & AWARDS

CIS Faculty Appreciation Award	2024	Eta Kappa Nu	2022, 2023, 2024, 2025
MCIT TA Award	2024	Tau Beta Pi	2023, 2024, 2025
CIS Senior Thesis	2024	Beta Gamma Sigma	2023, 2024, 2025
Wharton Research Scholars	2024	Dean's List	2022, 2023

TEACHING EXPERIENCE

Graduate Teaching Assistant, *Formal Verification (CS 4/5160)*
Cornell University. 26Sp. Supervised by Prof. Michael Clarkson.
An undergraduate/graduate course on formal verification and mechanization in Rocq. Topics include logic, programming language semantics, and verification of algorithms and data structures.

Graduate Teaching Assistant, *Programming Languages and Logics (CS 4/5110)*
Cornell University. 25Fa. Supervised by Prof. Nate Foster.
An undergraduate/graduate course on the theory, design, and implementation of programming languages. Topics include operational semantics, denotational semantics, axiomatic semantics, program logics, lambda calculus, dependent type systems, type safety, and logical relations.

Teaching Assistant, *Compilers and Interpreters (CIS 5521)*
University of Pennsylvania. 25Sp. Supervised by Prof. Steve Zdancewic.
A graduate course on compilers in OCaml. Topics include lexical analysis, grammars, parsing, intermediate representations, syntax- directed translation, code generation, type checking, dataflow and control-flow analyses, and optimizations.

Teaching Assistant, *Advanced Programming (CIS 5520)*
University of Pennsylvania. 24Fa. Supervised by Prof. Stephanie Weirich.
A graduate course on Haskell and programming paradigms. Topics include functional programming, type class, functor, applicative, monad, monad transformers, dependent type, refinement type. Taught 4 lectures. Mentored 4 project groups.

Head Teaching Assistant, *Analysis of Algorithms (CIS 5020 / CIT 5960 / CIS 3200)*
University of Pennsylvania. 22Su, 22Fa, 23Sp, 23Su, 23Fa, 24Su. Supervised by Prof. Sanjeev Khanna.
An undergraduate/graduate course on the paradigms for design and analysis of algorithms. Topics include dynamic programming, graph algorithms, flow and combinatorial optimization algorithms, linear programming, randomization, intractability, and approximation algorithms.

Head Teaching Assistant, *Theory of Networks (NETS 3120)*
University of Pennsylvania. 24Sp. Supervised by Prof. Victor Preciado.
An undergraduate course on network and data science. Topics include graph theory, social network analysis, graph machine learning, knowledge graphs, and graph generation.

Teaching Assistant, *Software Foundations (CIS 5000)*
University of Pennsylvania. 23Fa. Supervised by Prof. Benjamin Pierce.
A graduate course on basic concepts and techniques of programming languages. Topics include Rocq Proof Assistant, constructive logic, Hoare Logic, lambda calculus, type system, type safety, and mutable reference. Taught 1 lecture about maps in mechanization.

STEM Tutor, *The Weingarten Center*
University of Pennsylvania. 21Sp, 21Su, 21Fa, 22Sp, 22Su.
Taught Programming Languages and Techniques I (CIS 1200), Programming Languages and Techniques II (CIS 1210), Mathematical Foundations of Computer Science (CIS 1600), Introduction to Algorithms (CIS 3200), Calculus II (MATH 1410), Calculus III (MATH 2400).

RELEVANT COURSEWORK

Computer Science (Graduate): Kleene Algebra (CS 6861); Category Theory (CS 6117); Compilers (CIS 7000); Friendly Logics (CIS 6820); Randomized and Sublinear Algorithms (CIS 6770); Separation Logics (CIS 6700); Type Theory, Lambda Calculus, and Effect (CIS 6700); Formal Verification (CIS 5990); Computer Organization and Design (CIS 5710); Networked Systems (CIS 5530); Advanced Programming (CIS 5520); Database and Information Systems (CIS 5500); Operating Systems (CIS 5480); Software Analysis (CIS 5470); Big Data Analytics (CIS 545); Machine Learning (CIS 5200); Theory of Computation (CIS 5110); Software Foundations in Rocq (CIS 5000); Optimization (ESE 504); Applied Econometrics (STAT 5200).

Computer Science (Undergraduate): Algorithmic Game Theory (NETS 412); Network Theory (NETS 3120); Distributed Cloud Programming (NETS 212); Algorithms (CIS 320); Intro to Computer Systems (CIS 2400); Discrete Math (CIS 160); Foundations of Data Science (ESE 3050); Bayesian Statistics (STAT 4420); Statistical Inference (STAT 431); Probability (STAT 430); Linear Algebra (MATH 313 / MATH 240); Multivariable Calculus (MATH 114).

Business: Economics; Corporate Finance; Monetary Economics; Accounting; Cost Analysis; Forensic Analytics; Business Analytics; Negotiations; Technical Analysis; Game Theory.

COMMUNITY INVOLVEMENT

Artifact Evaluation Committee, PLDI 2026	Jan 2026	Student Volunteer, POPL 2025	Jan 2025
Artifact Evaluation Committee, ATVA 2026	Jul 2026	Student Volunteer, SPLASH 2024	Oct 2024
Artifact Evaluation Committee, POPL 2026	Nov 2025	Student Volunteer, OPLSS 2024	Jun 2024

OTHER EXPERIENCE

Administrator & Lecturer, *Gateways Summer Program*, supervised by Steve Downes — Feb 2025 – Jul 2025
Tabor Academy

- Co-organized a summer program hosting 65 students from Shenzhen Nanshan Foreign Language School across two eight-day sessions, providing exposure to U.S. boarding high school life and STEM courses, including marine science, computer science, and global economies.
- Taught seminars on business negotiation using interactive group activities, covering concepts such as fixed-pie perception, deal breakers, bargaining zones, and practical negotiation strategies.

Research Assistant, *The Knowledge Puzzle Project*, supervised by Prof. Andrew Carton — Feb 2021 – Sep 2021
University of Pennsylvania, The Wharton School

- Conducted research on automating knowledge database construction through PDF data extraction.

Research Intern (Volunteer), *Cybersecurity & Legislation* Jun 2019 – Aug 2019

Massachusetts State House, House Committee on Technology and Intergovernmental Affairs

- Researched and submitted a report for the Committee hearing regarding telehealth security, analyzing telehealth cybersecurity strategies at the federal and state levels, and suggesting potential improvements for legislation.
- Researched and wrote proposals regarding solutions to the cybersecurity workforce shortage in Massachusetts.